20 (a) Find the reciprocal of 20

Reciprocal = 1 ÷ 20 = 0.05

1 mark
Its product with 20 is 1.

0.05

(1)

(b) Work out the value of $\dfrac{3^{220}}{3^{190} \times 3^{28}}$

$3^{190} \times 3^{28} = 3^{218}$

$3^{220} \div 3^{218} = 3^2 = 9$

2 marks total
Add powers in the denominator; subtract when dividing.

9

(2)

Asma was asked to write $(5^{60})^3$ as a power of 5

Here is her working.

$(5^{60})^3 = 5^{60+3}$

$= 5^{63}$

(c) Is Asma's working correct?
You must give a reason for your answer.

No. Multiply the powers: 60 × 3 = 180.

The correct result is 5^{180}.

1 mark
A power of a power multiplies indices; it does not add them.

(1)

(Total for Question 20 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
AREA